AI-based IGRT with 2D Image-Guided Dose Tracking of Daily Cone-Beam CTs for Assessing Prostate Position, Rectum, and Bladder Filling in Definitive Radiotherapy of Prostate Carcinoma

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Abstract

Background: A method for the automatic detection of the bladder and rectum in daily Cone-Beam CTs (kV-CBCTs) for image-guided radiotherapy (IGRT) is presented.
Methods: Medial sagittal CT slices were segmented using a neural network (AI) to assess organ position and filling states of the rectum and bladder during definitive radiotherapy of prostate cancer.
Results: The model was trained and validated with data from 131 patients, with 5 kV-CBCTs per patient. The results show high segmentation performance with mean Dice scores of approximately 0.98 for the bladder and prostate PTV, and 0.95 for the rectum.
Summary: The AI-based method can effectively automate the verification of organ position and filling, offering potential to improve treatment quality and efficiency in IGRT.
Keywords: Segmentation, Prostate, Organs at Risk, Deep Learning, Medical Imaging, Cone-Beam CT, CBCT, Radiotherapy, Image-Guided Radiotherapy, Artificial Intelligence

1 Introduction

In the definitive irradiation of low- and intermediate-risk prostate carcinomas, percutaneous radiotherapy can be applied in standard fractionation, moderate hypofractionation, or ultrahypofractionation (???). A total dose is administered in daily fractions.

Interfractional variability due to changes in filling states or organ movements in the pelvic region can expose organs at risk (OARs) such as the bladder and rectum to undesirably high doses, potentially increasing acute or chronic toxicity (?).

To minimize these toxicities, a central step in modern radiotherapy is the verification of the exact position of the tumor and surrounding OARs before each fraction using image-guided radiotherapy (IGRT) (??). Various IGRT methods exist, e.g., Cone-Beam CTs (CBCT) or fiducials (gold markers) implanted in the prostate combined with planar X-ray images (?). However, verification of OAR filling states is only possible with CBCT. This allows varying fillings and movements to be accounted for and corrected, such as an inadequately filled bladder or the state of the highly variable rectum, to ensure compliance with dose constraints like $V_{65} < 50\%$ for the bladder (?).

Online-adaptive radiotherapy adjusts these changes daily but requires significant time and personnel effort (15–35 minutes per fraction with a specialist physician and medical physicist), making it infeasible in high-throughput clinics (?). Frequently, only an offline review is performed, where deviations are detected retrospectively.

Inspired by advances in medical image segmentation, we develop an AI-based method for partial automation of online-adaptive therapy to more efficiently monitor the position and filling of OARs in daily CBCT scans and improve treatment quality.

2 Methods

Between July 1, 2019, and January 31, 2024, 131 patients with low- to intermediate-risk prostate carcinoma were retrospectively selected who received definitive, moderately hypofractionated radiotherapy (25 fractions with simultaneous integrated boost, 58–68 Gy), based on the schema of (?). Inclusion criteria were cT1a–cT2b, PSA $\leq 20\,\text{ng/ml}$, and Gleason Score ≤ 7; patients with high-risk profiles or distant metastases were excluded. Written consent for anonymous data use was obtained.

For each patient, a planning CT (PL-CT) and 5 kV-CBCTs were anonymously extracted from the RayStation system (versions 10B–12A) (?). Structures (prostate PTV, bladder, rectum) were manually contoured by a medical physicist and reviewed by a specialist physician. A custom Python program (*sqlite_-dicom2.py*) (?) converted DICOM data into volumes, extracted three medial sagittal 2D slices per kV-CBCT—central through the symphysis and one each to the left and right—and generated binary masks using *scipy.ndimage.ConvexHull*. The outputs (patient ID, DICOM series, slices, masks, slice number) were stored in an SQLite database. A second program (*sqlite_viewer.py*) (?) enabled visual inspection and data cleaning.

Images were preprocessed (exclusion of unusual sizes, normalization of pixel intensity). In total, approximately 2100 images per organ were generated (3 slices $\times$ 5 kV-CBCTs $\times$ 131 patients), with about 1700 for training and 400 for validation, using patient-disjoint datasets. The training images were augmented, resulting in approximately 9000 images to improve model robustness (???). The augmentation techniques used are listed in Table ??.

Augmentation	**Parameters**
Rotation	$-45°$ to $45°$
Mirroring	None
Noise	Gaussian noise, variance: 0.001
Distortion	Shear: -0.1 to 0.1

Table 1: Augmentation techniques to improve model robustness.

A U-Net-based Convolutional Neural Network (CNN) ? was trained with PyTorch to segment kV-CBCT images (see Fig. ??). The model processes midsagittal CT slices and consists of an encoder-decoder architecture with skip connections, combining high-resolution features from the encoder with upsampled decoder outputs to obtain precise organ boundaries, such as those of the bladder. The contracting path (encoder) comprises four blocks, each with two 3×3 convolutions with ReLU activation and a 2×2 max-pooling; the convolutions extract local features like edges or textures, with padding used to preserve spatial dimensions within blocks before max-pooling halves the dimensions stepwise. The number of feature channels (e.g., 64, 128, 256 in the encoder) indicates the number of learned feature maps, reflecting the network's ability to extract complex features. The expansive path (decoder) uses transposed convolutions for upsampling to restore the original dimensions, producing a binary segmentation mask after a final inference step.

Training was performed on an AMD RX 7900 XTX GPU with 24 GB VRAM, accelerated by the ROCm stack. A learning rate of 10^{-4} was used with the Adam optimizer, and the model converged after 25 epochs. The loss function was binary cross-entropy with logits (BCEWithLogitsLoss). Only CBCT data were used to ensure consistency ?. Training data were organized in a custom `ImageDataset` as PyTorch tensors, stacking Pandas Series into NumPy arrays and then converting them to tensors. A `DataLoader` (batch size 32, 4 workers, `pin_memory=True`) was used to load data efficiently. During training, images and masks were moved to the GPU in batches, with the loss computed, backpropagated, and optimized after each batch. The training time was approximately 1 hour per organ.

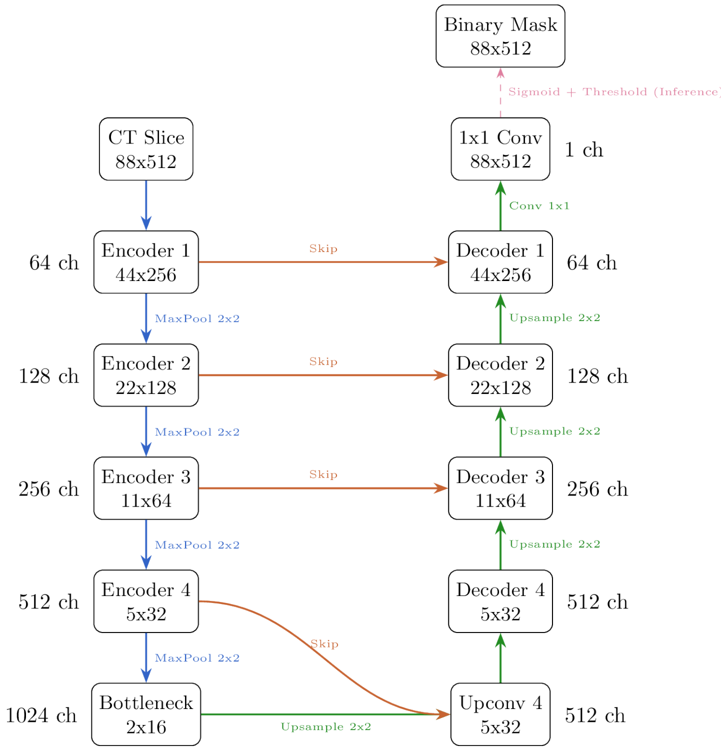


Figure 1: Schematic representation of the U-Net architecture for segmenting kV-CBCT images. Size of feature maps at each step specified in pixels. Color coding: Blue: Encoder path, Green: Decoder path, Orange: Skip connections. ch: Feature channels. Dashed line indicates inference step.

The bladder volume was roughly estimated by measuring the height and width of the bladder mask in the midsagittal plane and calculating the volume of a prolate spheroid. For the rectum, the volume of a cylinder was used for estimation.

The process of data preparation, model training, and validation can be viewed at **?**, in the Jupyter notebooks `data_prep_and_training_`X`.ipynb`, where X is replaced by the respective organ.

3 Results

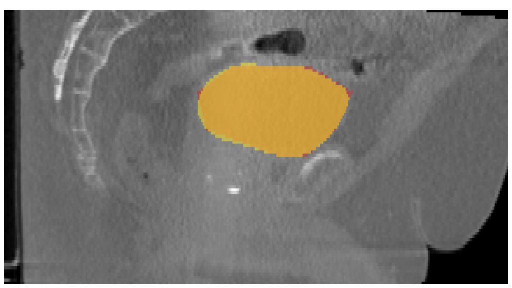
The U-Net model was tested on approximately 400 validation images per organ. The segmentation accuracy yielded the following Dice coefficients, averaged over all validation images, see Table **??**:

Organ	**Dice Coefficient**	**Standard Deviation**
Bladder	0.9815	± 0.0126
Prostate	0.9789	± 0.0092
Rectum	0.9552	± 0.1021

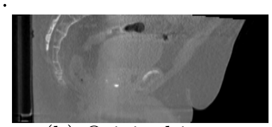
Table 2: Dice coefficients for organ segmentation.

The volumes of the bladder and rectum were estimated. These values can be compared with the volumes from the PL-CT to assess filling states.

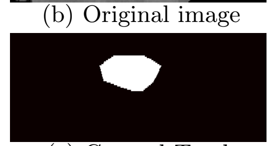
The manually created contours, reviewed by a specialist physician, served as the "ground truth." Model predictions were visually compared with the ground-truth data. Examples are shown in Figures **??**, **??**, and **??**.



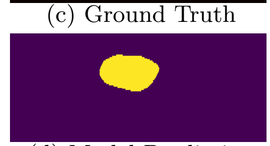
(a) Merged image of the bladder. Red: Ground Truth, Yellow: Prediction, Orange: Overlap



(b) Original image



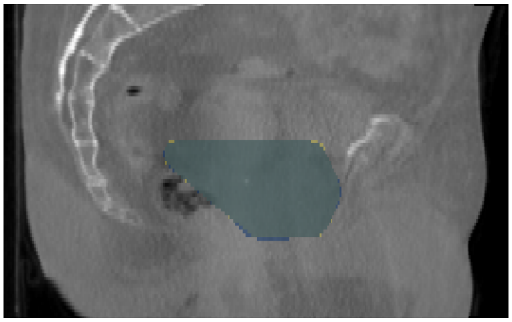
(c) Ground Truth



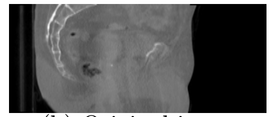
(d) Model Prediction

Figure 2: Example segmentation of the bladder.

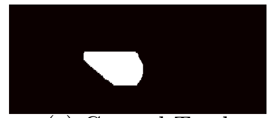
In Figure **??**, the segmentation visibly covers the bladder better than the ground truth, which appears jagged due to segmentation artifacts. The model prediction is smoother and fully encompasses the bladder. The bladder filling in this example was estimated at 150 ml.



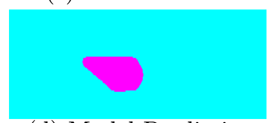
(a) Merged image of the prostate PTV. Yellow: Ground Truth, Dark Blue: Prediction, Cyan: Overlap



(b) Original image



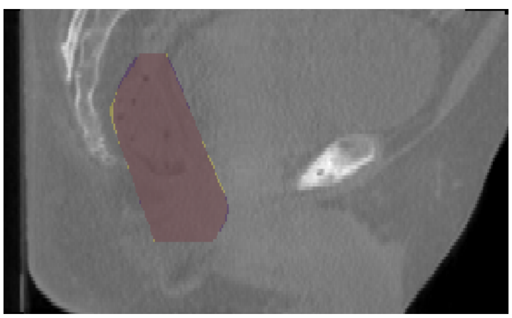
(c) Ground Truth



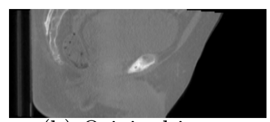
(d) Model Prediction

Figure 3: Example segmentation of the prostate PTV.

The model prediction in Figure **??** covers the prostate PTV very well. This is of academic interest since the PTV lacks clearly defined anatomical boundaries as is the case with the bladder. In practice, the actual PTV (rather than an estimation) can be overlaid on the CBCT to estimate dosing to the organs at risk.

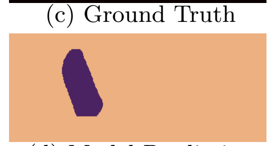


(a) Merged image of the rectum. Yellow: Ground Truth, Dark Brown: Prediction, Light Brown: Overlap



(b) Original image

(c) Ground Truth



(d) Model Prediction

Figure 4: Example segmentation of the rectum.

The results for the rectum in Figure **??** are also satisfactory but less precise than for the bladder and prostate PTV. This is reflected in the lower Dice coefficient and higher standard deviation. The rectum volume in this example was estimated at 300 ml.

4 Discussion

The method enables precise segmentation of the bladder and rectum. The Dice value of 0.9815 for the bladder and 0.9789 for the prostate PTV provides reliable results for daily use in IGRT. The value of 0.9552 ± 0.1021 for the rectum is also acceptable, particularly in conjunction with visualization, with the higher standard deviation reflecting its greater variability (**?**).

The trained models can be executed on a standard computer in a few seconds (near real-time), enabling integration into clinical practice, e.g., as a plugin for the RayStation or Aria system, making daily organ position verification feasible.

In the future, AI-based segmentation could replace the use of fiducials (e.g., gold markers for IGRT) in certain cases, reducing treatment costs and time as well as avoiding risks associated with invasive procedures.

Extending the model to the entire kV-CBCT volume would enable 3D segmentation of organs at risk, increasing the method's accuracy and robustness. Direct measurement of organ volumes would be particularly useful. However, this would likely require a larger data base and more computational power.

Limitations of the method include dependence on consistent kV-CBCT parameters and inaccuracy due to information loss from dimension reduction to 2D. Future work could employ transfer learning (TL) or advanced augmentation to improve robustness across variable devices.

The 2D U-Net model developed in this study achieves Dice values surpassing the results of Isaksson et al. (**?**). Our models achieve mean Dice values of 0.98 for the bladder and prostate PTV and 0.95 for the rectum—values indicating very high agreement with reference contours. In comparison, Isaksson et al. investigated six segmentation methods, including a 2D U-Net, on T2-weighted MRI images from 100 patients. Their best model, a customized EfficientDet architecture, achieved a mean Dice value of 0.914 for the entire 3D prostate volume, while other models, including the 2D U-Net, yielded values between 0.855 and 0.907. This performance difference can be attributed to several factors. First, we used kV-CBCT data, unlike Isaksson's T2 MRI sequences, which may have allowed better adaptation to specific image characteristics. Second, our model is based on a larger training dataset—approximately 9000 augmented images from 1700 originals from 131 patients—compared to Isaksson's 100 patients (also augmented). Third, our focus on 2D slices (medial sagittal planes) simplifies segmentation compared to Isaksson's 3D volume approach, which is more complex and demands a higher data base. Crucially, our Dice values are based on a planar 2D calculation per slice, whereas Isaksson et al. computed the Dice coefficient over the entire 3D volume, limiting comparability since 3D calculations can yield lower values due to spatial variations.

5 Conclusion

The AI-based method effectively automates the verification of prostate position and organ filling, offering potential to improve treatment quality and efficiency in IGRT. The high Dice coefficients underscore the reliable segmentation of organs, supporting its use in clinical practice.

6 References

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